



Faculty of Computer Applications

**BCA Sem-1
(AY-2026-27)**

Lab Book

**Sub Code: 05BC3103
(Programming Practices: 1)**

Roll No :	92600527073
Full Name :	Siddhi.S.Gangani

GitHub Link | <https://github.com/siddhigangani9>

**Sub. Faculty Name
Prof. Vipulkumar Bambhaniya**

Index

Unit : 1	
Program No	Definition
	Draw Algorithm and Flow chart for the following
	Definitions
1.	Print "Hello World".
2.	Takes 2 Values from user and Perform all Arithmetic Operations.
3.	Print numbers from 1 to 20.
4.	Input 3 values from user and find out its average.
5.	Calculate the square and cube of a number.
6.	Interchange the value of two numbers.
7.	Find largest out of three numbers.
8.	Find out fibonacci series up to given n values.
9.	Find out number is Odd or Even.
10.	Find out sum of digits of a given number.

	Unit: 2
Program No	Definition
1.	Write a program to print "Hello World".
2.	Write a program that takes 2 Values from user and perform all Arithmetic Operation.
3.	Write a program that takes 3 values for principle amount, Rate of Interest and no of years and find out simple Interest.
4.	Write a program that input value from user and find out area of circle.
5.	Write a program that input two values from user and find out area of rectangle.
6.	Write a program that input 3 values from user and find out its average.
7.	Write a program to calculate the square and cube of a number.
8.	Write a program Interchange the value of two numbers using third variable.
9.	Write a program Interchange the value of two numbers without using third variable.
10.	Write a program that input quantity, price, discount and display final amount.

1.print"Hello World"

```
#include <stdio.h>
#include <conio.h>

void main()
{
    clrscr();
    printf("Hello World\n");
    getch();
}
```

Output :

A screenshot of a Turbo C++ console window. The window has a black background with white text. The first line shows the command prompt path 'C:\TURBOC3\BIN>' followed by the command 'TC'. The second line shows the output 'Hello World'.

2. Takes 2 Values from user and Perform All Arithmetic Operations.

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a,b,sum,sub,multi;
    clrscr();
    printf("--Calculator--");
    printf("\nEnter a number:");
    scanf("%d",&a);
    printf("\nEnter another number:");
    scanf("%d",&b);
    sum=a+b;
    printf("\nsum of %d and %d is %d",a,b,sum);
    sub=a-b;
    printf("\nsubtraction of %d and %d is %d",a,b,sub);
    multi=a*b;
    printf("\nmultiplication of %d and %d is %d",a,b,multi);
    if(b==0){
        printf("\nDivision by zero error");
    }
    else{
        printf("\ndivision of %d and %d is %d",a,b,a/b);
        printf("\nremainder of %d and %d is %d",a,b,a%b);
    }
    getch();
}
```

Output :

```
--Calculator--  
Enter a number:5  
  
Enter another number:5  
  
sum of 5 and 5 is 10  
subtraction of 5 and 5 is 0  
multiplication of 5 and 5 is 25  
division of 5 and 5 is 1  
remainder of 5 and 5 is 0_
```

3. Print numbers from 1 to 20.

```
#include<stdio.h>

#include<conio.h>

void main()
{
    int a;

    clrscr();

    for (a=1; a<=20; a++)
    {
        printf("\n%d",a);
    }

    getch();
}
```

Output :

```
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
-
```

4. Input 3 values from user and find out its average.

```
#include<stdio.h>

#include<conio.h>

void main()

{

    int a,b,c,sum,avg;

    clrscr();

    printf("Enter a number:");

    scanf("%d",&a);

    printf("Enter another number:");

    scanf("%d",&b);

    printf("Enter another number:");

    scanf("%d",&c);

    sum=a+b+c;

    avg=sum/3;

    printf("Average is %d",avg);

    getch();

}
```

Output :

```
Enter a number:12  
Enter another number:15  
Enter another number:20  
Average is 15_
```

5. Calculate the square and cube of a number.

```
#include<stdio.h>

#include<conio.h>

void main()
{
    int a,sq,cu;

    clrscr();

    printf("Enter a number:");

    scanf("%d",&a);

    sq=a*a;

    cu=a*a*a;

    printf("\nsquare of %d is %d",a,sq);

    printf("\ncube of %d is %d",a,cu);

    getch();
}
```

Output :

```
Enter a number:2
```

```
square of 2 is 4
```

```
cube of 2 is 8
```

6. Interchange the value of two numbers.

```
#include<stdio.h>

#include<conio.h>

void main()

{

    int a,b;

    clrscr();

    printf("Enter a number:");

    scanf("%d",&a);

    printf("Enter a number:");

    scanf("%d",&b);

    printf("Numbers before swap");

    printf("\nA is %d",a);

    printf("\nB is %d",b);

    a=a+b;//5+4=9

    b=a-b;//9-4=5

    a=a-b;//9-5=4

    printf("\nNumbers after swap");

    printf("\nA is %d",a);

    printf("\nB is %d",b);
```

```
    getch();  
}
```

Output :

A screenshot of a terminal window with a black background and white text. The text shows the execution of a program where two numbers, 5 and 4, are entered and then swapped. The output is as follows:

```
Enter a number:5  
Enter a number:4  
Numbers before swap  
A is 5  
B is 4  
Numbers after swap  
A is 4  
B is 5
```

Enter a number:5
Enter a number:4
Numbers before swap
A is 5
B is 4
Numbers after swap
A is 4
B is 5

7. Find largest out of three numbers.

```
#include<stdio.h>

#include<conio.h>

void main()
{
    int a,b,c;

    clrscr();

    printf("Enter a number:");

    scanf("%d",&a);

    printf("Enter another number:");

    scanf("%d",&b);

    printf("Enter another number:");

    scanf("%d",&c);

    if(a>b){

        if(a>c){

            printf("A is max");

        }

        else{

            printf("C is max");

        }

    }

    else{

        if(b>c){

            printf("B is max");

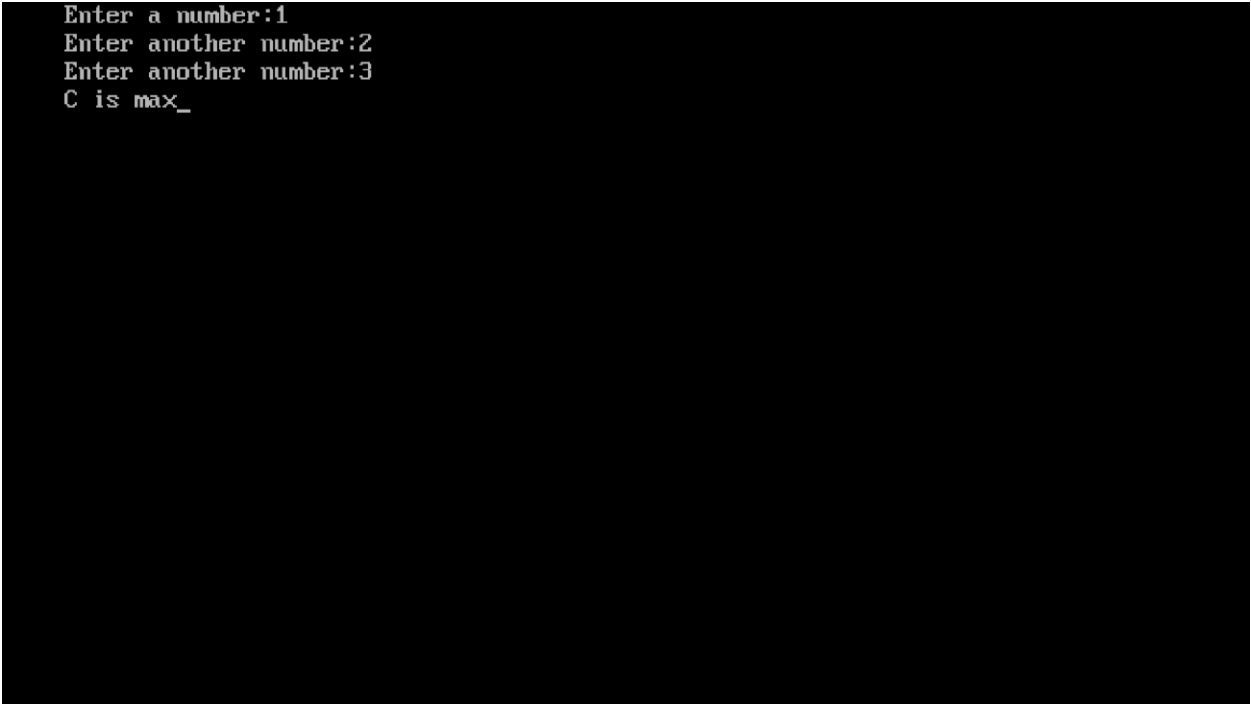
        }

    }

}
```

```
else{  
    printf("C is max");  
}  
}  
  
getch();  
}
```

Output :

A screenshot of a terminal window with a black background and white text. The text shows the output of a C program where three numbers are entered, and the maximum is identified. The output is: Enter a number:1, Enter another number:2, Enter another number:3, and C is max_.

```
Enter a number:1  
Enter another number:2  
Enter another number:3  
C is max_
```

8. Find out fibonacci series up to given n values.

```
#include<stdio.h>

#include<conio.h>

void main()

{

    int n,next,first=0,i,second=1;

    clrscr();

    printf("Enter a number:");

    scanf("%d",&n);

    for(i =1; i <= n; i++){

        next=first+second;;

        first=second;

        second=next;

        printf("\n%d",next);

    }

    getch();

}
```

Output :

```
Enter a number:9
```

```
1
2
3
5
8
13
21
34
55_
```

9. Find out number is Odd or Even.

```
#include<stdio.h>

#include<conio.h>

void main()

{

    int a;

    clrscr();

    printf("Enter a number:");

    scanf("%d",&a);

    if (a%2==0)

    {

        printf("It's even");

    }

    else

    {

        printf("It's odd");

    }

    getch();

}
```

Output :

```
Enter a number:4  
It's even_
```

10. Find out sum of digits of a given number.

```
#include<stdio.h>
```

```
#include<conio.h>
```

```
void main()
```

```
{
```

```
int num1,num2,sum;
```

```
clrscr();
```

```
printf("enter number1:");
```

```
scanf("%d",&num1);
```

```
printf("enter number2:");
```

```
scanf("%d",&num2);
```

```
sum=num1+num2;
```

```
printf("sum of %d and %d is: %d",num1,num2,sum);
```

```
getch();
```

```
}
```

Output :

```
enter number1:5  
enter number2:6  
sum of 5 and 6 is: 11_
```

Unit-2

1. Write a program to print "Hello World".

```
#include <stdio.h>
#include <conio.h>

void main()
{
    clrscr();
    printf("Hello World\n");
    getch();
}
```

Output :

A screenshot of a Turbo C++ compiler window. The title bar reads 'C:\TURBOC3\BIN>TC'. The main window area is black with white text. The first line shows the command prompt 'C:\TURBOC3\BIN>TC'. The second line shows the output 'Hello World'.

2. Write a program that takes 2 Values from user and perform all Arithmetic Operation.

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a,b,sum,sub,multi;
    clrscr();
    printf("--Calculator--");
    printf("\nEnter a number:");
    scanf("%d",&a);
    printf("\nEnter another number:");
    scanf("%d",&b);
    sum=a+b;
    printf("\nsum of %d and %d is %d",a,b,sum);
    sub=a-b;
    printf("\nsubtraction of %d and %d is %d",a,b,sub);
    multi=a*b;
    printf("\nmultiplication of %d and %d is %d",a,b,multi);
    if(b==0){
        printf("\nDivision by zero error");
    }
    else{
        printf("\ndivision of %d and %d is %d",a,b,a/b);
        printf("\nremainder of %d and %d is %d",a,b,a%b);
    }
    getch();
}
```

Output :

```
--Calculator--  
Enter a number:5  
  
Enter another number:0  
  
sum of 5 and 0 is 5  
subtraction of 5 and 0 is 5  
multiplication of 5 and 0 is 0  
Division by zero error_
```

3. Write a program that takes 3 values for principle amount, Rate of Interest and no of years and find out simple Interest.

```
#include<stdio.h>

#include<conio.h>

void main()

{

    float amt,interest,years,eq;

    clrscr();

    printf("Enter principle amount:");

    scanf("%f",&amt);

    printf("Enter rate of interest:");

    scanf("%f",&interest);

    printf("Enter no of years:");

    scanf("%f",&years);

    eq=(amt*interest*years)/100;

    printf("\nSimple interest is %f",eq);

    getch();

}
```

Output :

```
Enter principle amount:10000
Enter rate of interest:12
Enter no of years:0.5

Simple interest is 600.000000
```

4. Write a program that input value from user and find out area of circle.

```
#include<stdio.h>

#include<conio.h>

void main()

{

    float pi=3.14,r,eq;

    clrscr();

    printf("Enter a number:");

    scanf("%f",&r);

    eq=pi*r*r;

    printf("Area of circle is %f",eq);

    getch();

}
```

Output :

```
Enter a number:5  
Area of circle is 78.500000
```

5. Write a program that input two values from user and find out area of rectangle.

```
#include<stdio.h>

#include<conio.h>

void main()

{

    int a,b,eq;

    clrscr();

    printf("Enter height of rectangle:");

    scanf("%d",&a);

    printf("Enter width of rectangle:");

    scanf("%d",&b);

    eq=a*b;

    printf("Area of rectangle %d",eq);

    getch();

}
```

Output :

```
Enter height of rectangle:5  
Enter width of rectangle:7  
Area of rectangle 35
```

6. Write a program that input 3 values from user and find out its average.

```
#include<stdio.h>

#include<conio.h>

void main()
{
    int a,b,c,sum,avg;

    clrscr();

    printf("Enter a number:");

    scanf("%d",&a);

    printf("Enter another number:");

    scanf("%d",&b);

    printf("Enter another number:");

    scanf("%d",&c);

    sum=a+b+c;

    avg=sum/3;

    printf("Average is %d",avg);

    getch();
}
```

Output :

```
Enter a number:34  
Enter another number:45  
Enter another number:56  
Average is 45
```

7. Write a program to calculate the square and cube of a number.

```
#include<stdio.h>

#include<conio.h>

void main()

{

    int a,sq,cu;

    clrscr();

    printf("Enter a number:");

    scanf("%d",&a);

    sq=a*a;

    cu=a*a*a;

    printf("\nSquare of %d is %d",a,sq);

    printf("\nCube of %d is %d",a,cu);

    getch();

}
```

Output :

```
Enter a number:4
```

```
Square of 4 is 16
```

```
Cube of 4 is 64_
```

8. Write a program Interchange the value of two numbers using third variable.

```
#include<stdio.h>

#include<conio.h>

void main()

{

    int a,b,temp;

    clrscr();

    printf("\nEnter a number:");

    scanf("%d",&a);

    printf("\nEnter another number:");

    scanf("%d",&b);

    printf("\nBefore Swapping");

    printf("\nA is %d",a);

    printf("\nB is %d",b);

    temp=a;

    a=b;

    b=temp;

    printf("\nAfter swapping");

    printf("\nA is %d",a);

    printf("\nB is %d",b);
```

```
    getch();  
}
```

Output :

```
Enter a number:5  
Enter another number:6  
  
Before Swapping  
A is 5  
B is 6  
After swapping  
A is 6  
B is 5
```

9. Write a program Interchange the value of two numbers without using third variable.

```
#include<stdio.h>

#include<conio.h>

void main()

{

    int a,b;

    clrscr();

    printf("Enter a number:");

    scanf("%d",&a);

    printf("Enter a number:");

    scanf("%d",&b);

    printf("Numbers before swap");

    printf("\nA is %d",a);

    printf("\nB is %d",b);

    a=a+b;//10+20=30

    b=a-b;//30-20=10

    a=a-b;//30-10=20

    printf("\nNumbers after swap");

    printf("\nA is %d",a);

    printf("\nB is %d",b);
```

```
    getch();  
}
```

Output :

```
Enter a number:10  
Enter a number:20  
Numbers before swap  
A is 10  
B is 20  
Numbers after swap  
A is 20  
B is 10
```

10. Write a program that input quantity, price, discount and display final amount.

```
#include<stdio.h>

#include<conio.h>

void main()

{

    float pr,qu,disc,totalbef,discamt,finalamt;

    clrscr();

    printf("Enter price:");

    scanf("%f",&pr);

    printf("Enter quantity:");

    scanf("%f",&qu);

    printf("Enter discount percent:");

    scanf("%f",&disc);

    totalbef=pr*qu;

    discamt=(totalbef*disc)/100.0;

    finalamt=totalbef-discamt;

    printf("-----bill-----");

    printf("\nTotal amount before %f",totalbef);

    printf("\nDiscount given: %f",discamt);

    printf("\nFinal amount: %f",finalamt);
```

```
    getch();  
}
```

Output :

```
Enter price:5  
Enter quantity:200  
Enter discount percent:10  
-----bill-----  
Total amount before 1000.000000  
Discount given: 100.000000  
Final amount: 900.000000
```